

A Mechanism-Informed Gene Signature for Regenerative vs. Fibrotic Tendon Healing

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1. Biological Problem and Significance

Tendons heal via one of two outcomes — regeneration, restoring fibrillar architecture, or fibrosis, producing permanent scar — yet the determinants of this dichotomy are poorly understood. Harvey et al. (2019) resolved the cellular basis using scRNA-seq: two co-resident PDGFR α + progenitor populations occupy the tendon sheath, with Tppp3+ Tendon Stem/Progenitor Cells (TSPCs) driving regeneration and Tppp3- Tendon Fibro/Adipogenic Progenitors (T-FAPs) driving fibrosis. Cherief et al. (2023) identified TrkA/NGF-mediated reinnervation as a key upstream switch: surgical denervation blocks TSPC expansion and shifts healing toward fibrosis, implicating NGF, TGF β (Kaji et al., 2020), and PDGF as candidate fate-control signals. Despite this progress, the TF programs executing the TSPC vs. T-FAP fate decision downstream of these signals are uncharacterized, and no validated computational tool exists to objectively quantify healing trajectory in a given sample.

This project addresses both gaps with two linked objectives: (1) map TF regulatory programs governing TSPC/T-FAP fate specification via TF network inference, trajectory modeling, and cell-cell communication analysis of the Cherief 2023 innervated/denervated comparison; (2) compress mechanism-derived gene programs into a validated minimal panel classifying healing outcome, with output as a continuous healing index score. TF regulons from objective (1) provide a biologically grounded prior for feature selection in objective (2), yielding a mechanism-informed rather than purely data-driven signature.

2. Hypothesis

NGF/TGF β /PDGF signals from the innervated microenvironment selectively activate TSPC-specific TF regulons to bias PDGFR α + progenitor fate toward regeneration; denervation suppresses this axis, permitting T-FAP-mediated fibrosis by default. pySCENIC (Van de Sande et al., 2020) will identify fate-specific TF programs on TSPC/T-FAP sub-clusters, and LIANA (Dimitrov et al., 2022) will determine which signaling axes are most disrupted under denervation. Secondly, a 10–30 gene subset of these TF programs is predicted to distinguish regenerative from fibrotic healing across anatomically distinct tendon datasets (held-out AUROC > 0.85).

3. Data Description

The primary dataset (Cherief 2023, GEO: GSE244921) profiles mouse Achilles tendon in innervated (TrkAWT, n=3) vs. denervated (TrkAF592A, n=3) conditions, yielding 22,615 QC-passed cells; it is downloaded and fully preprocessed, and also includes human tendon biopsies for cross-species validation. The secondary dataset (Harvey 2019, SRA: PRJNA506218) profiles uninjured adult mouse patellar tendon

(~2,491 cells, 8 clusters) and is the only published dataset with validated TSPC (Tppp3+, cluster 2) and T-FAP (cluster 3) annotations, making it the ground-truth source for classifier training labels. Raw FASTQs (23.2 GB) require Cell Ranger processing to generate a count matrix, which is a planned parallel task.

4. Preliminary Data

The Cherief 2023 dataset has been fully preprocessed: QC filters (min 200 genes/cell, max 7,500 genes/cell, <20% mitochondrial reads) retained 22,615/25,545 cells (88.5%), followed by normalization, log1p transformation, 2,000 HVG selection with condition-level batch correction, PCA (20 components), kNN (k=30), and Leiden clustering (resolution 0.5), yielding 17 clusters. The PDGFR α + stromal cluster (cluster 8, n=3,900 cells) was sub-clustered at resolution 0.2 into four annotated populations: TSPC (cluster 3; Tppp3 mean = 1.755, 94.5% expressing; Sfrp2 absent), T-FAP (cluster 1; Pi16 mean = 1.633, Sfrp2 mean = 1.347), a tenogenic progenitor (cluster 2; ECM markers Thbs4/Kera/Coll1a1; enriched for innervated cells), and a mixed stromal population (cluster 0; imprinted genes Plag1/Mest/H19).

A notable finding is that unbiased DEG analysis identifies T-FAPs by complement cascade genes (C3, C4b, Serping1) — unreported in Harvey 2019 — suggesting a complement-mediated fibrosis mechanism directly testable via LIANA. The TSPC/T-FAP condition split is already visible in UMAP: the TSPC zone is enriched for TrkAWT cells and T-FAP zone for TrkAF592A cells, reproducing the core finding of Cherief et al. (2023).

5. Computational Methods

5.1 Preprocessing

Preprocessing follows the standard scanpy workflow (QC filtering, normalization, log1p, HVG selection with condition batch correction, scaling, PCA, kNN, UMAP/Leiden); full parameters are documented in Section 4.

5.2 Transcription Factor Network Inference

pySCENIC (Van de Sande et al., 2020) will infer TF regulons on TSPC/T-FAP sub-populations via GRNBoost2 co-expression modules, cisTarget motif enrichment pruning (JASPAR 2022), and AUCell per-cell scoring. TSPC- vs. T-FAP-specific regulons will be identified by Wilcoxon test on AUCell scores (BH-FDR < 0.05, $|\Delta\text{AUC}| > 0.10$).

5.3 Fate Trajectory Analysis

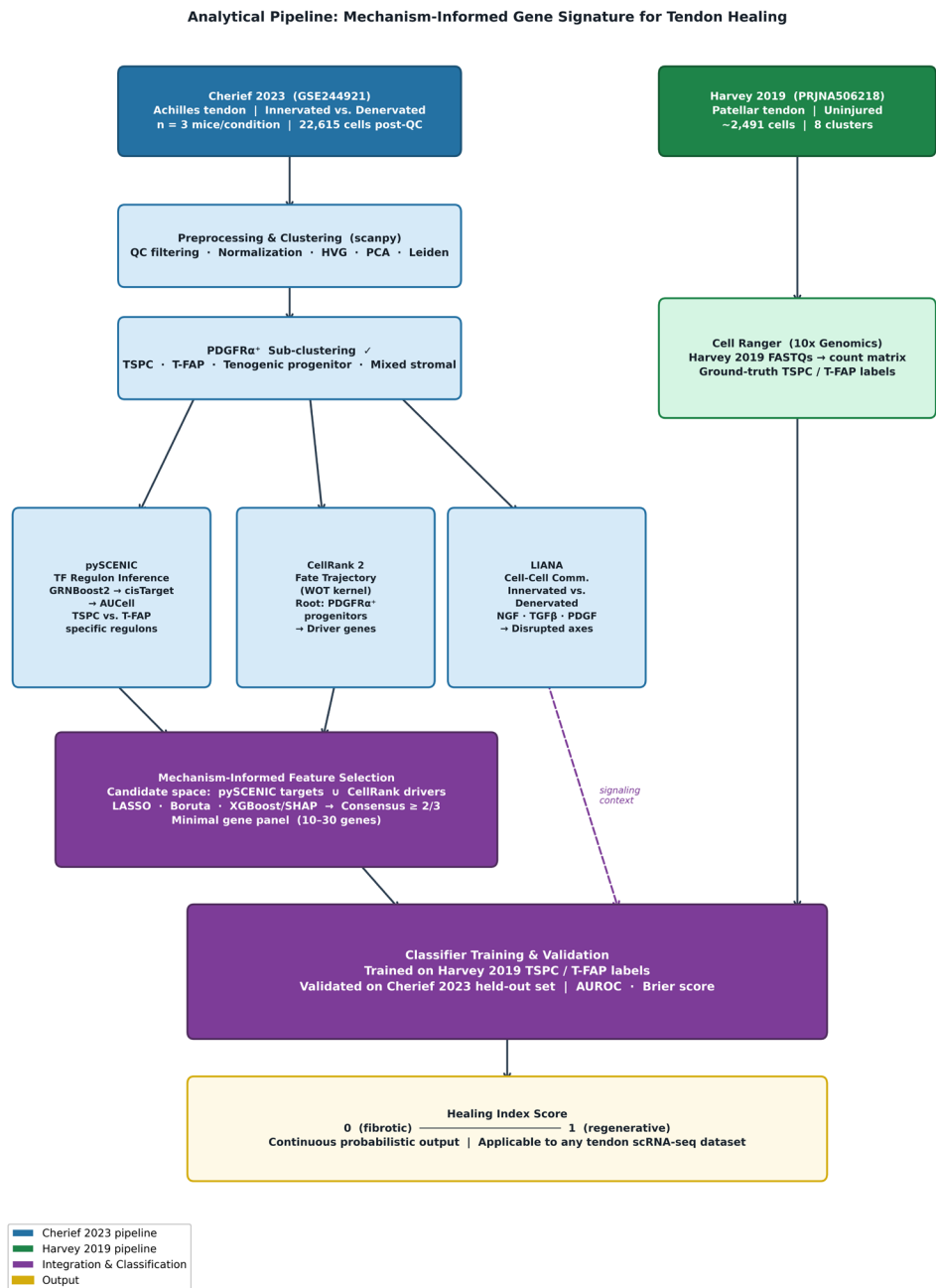
CellRank 2 (Weiler et al., 2024) with the Waddington Optimal Transport kernel will model trajectories from PDGFR α + progenitors (root) to TSPC and T-FAP terminal states. Trajectory driver genes will be cross-referenced with pySCENIC regulons to identify TFs active at fate-commitment branch points.

5.4 Cell-Cell Communication Analysis

LIANA (Dimitrov et al., 2022) will compare innervated vs. denervated ligand-receptor interactions, focusing on NGF→TrkA/Ntrk1, TGFβ→Tgfr1/2, and PDGF→Pdgfra axes with TSPC/T-FAP sub-clusters as receivers. Pseudobulk aggregation per mouse will be applied to reduce pseudo-replication.

5.5 Pipeline Summary

Sub-clustering of cluster 8 is complete. Cell Ranger (Harvey 2019) and pySCENIC both depend only on Cherief 2023 sub-cluster labels and will run in parallel. CellRank and LIANA are likewise independent of Harvey 2019. Feature selection requires pySCENIC and CellRank outputs; classifier training requires both feature selection and the Harvey 2019 count matrix.



5.6 Feature Selection and Classifier

Feature selection operates on the union of pySCENIC regulon targets and CellRank driver genes. Three parallel methods — LASSO (5-fold CV), Boruta, and XGBoost/SHAP — each nominate candidates; genes selected by ≥ 2 methods constitute the minimal panel. The classifier is trained on Harvey 2019 TSPC/T-FAP labels and evaluated on held-out Cherief 2023 data (AUROC, Brier score); output probability is interpreted directly as a continuous healing index score (0 = fibrotic, 1 = regenerative).

6. Expected Outcomes and Interpretation

Three deliverables are expected: (1) a TF regulatory circuit diagram linking fate-specific regulons to upstream NGF/TGF β /PDGF ligand-receptor pairs — the first mechanistic framework connecting innervation to tendon progenitor fate — and evaluating whether the T-FAP complement signature (C3, C4b, Serping1) represents a novel fibrosis mechanism; (2) a validated 10–30 gene panel (AUROC > 0.85) generalizable across two tendon anatomies and tissue states; (3) the healing index score, a continuous output designed to replace subjective histological scoring for future laboratory datasets.

7. Computational Reproducibility

Analyses will be conducted primarily in Python. An environment specification file will be provided upon project completion to document package versions and enable independent reproduction. The project repository will follow a standardized directory structure and will be uploaded to GitHub upon completion, excluding raw data files, which are independently retrievable via the GEO and SRA accession numbers documented in the project README.

8. Contingency Plans

Four failure modes with defined contingencies: (1) If pySCENIC regulons are dominated by generic stress TFs (e.g., Jun/Fos), pivot to SCENIC+ (Bravo González-Blas et al., 2023) using Arostegui et al. (2022) scATAC-seq data from Hic1+ mesenchymal progenitors as a chromatin accessibility prior. (2) If TSPC/T-FAP populations cannot be cleanly separated by clustering, replace discrete labels with a continuous score (per-cell Tppp3 minus Sfrp2 scaled expression). (3) If Cell Ranger fails on Harvey 2019 FASTQs, contact the Harvey lab for the count matrix; if unavailable, use leave-one-mouse-out cross-validation within Cherief 2023. (4) If classifier AUROC falls below 0.75, convert output to AUCell-based regulon activity scores as a surrogate healing index.

9. Critical Evaluation of Limitations

The most significant limitation is the cross-tissue, cross-condition design: Harvey 2019 (uninjured patellar tendon) provides training labels while Cherief 2023 (injured Achilles tendon) is the validation set, so the signature should be interpreted as a first-pass cross-tendon, cross-condition result rather than a universal classifier. Compounding this, Harvey 2019 provides only ~200–500 TSPC and T-FAP cells for training; overfitting risk is mitigated by restricting the feature search space to mechanism-derived candidates and requiring consensus across three independent selection methods.

Sub-clustering was performed on the global PCA embedding rather than a cluster 8-specific re-normalization, which may reduce sensitivity to subtle intra-cluster variation. Additionally, n=3 mice per condition means condition-specific findings (especially LIANA and CellRank) risk reflecting mouse-to-mouse noise; pseudobulk aggregation before testing will address this where applicable.

The T-FAP complement signature (C3, C4b, Serping1) has not been validated by orthogonal methods and may be Achilles-injury-specific; it will be presented as hypothesis-generating and flagged as requiring experimental validation beyond this project's scope.

10. Expected Timeline

Week	Dates	Milestone
1	May 29 – Jun 4	Cluster 8 sub-clustering complete; pySCENIC environment setup and TF database download
2	Jun 5 – Jun 11	pySCENIC GRNBoost2 + cisTarget + AUCell on TSPC and T-FAP sub-clusters
3	Jun 12 – Jun 18	CellRank WOT trajectory; branch-point TF identification
4	Jun 19 — Progress report	LIANA cell-cell communication (innervated vs. denervated); pseudobulk aggregation
5	Jun 26 – Jul 2	Mechanistic integration — regulatory circuit draft; Harvey 2019 Cell Ranger (parallel)
6	Jul 3 – Jul 9	Mechanism-informed feature selection: LASSO + Boruta + XGBoost
7	Jul 10 – Jul 16	Classifier training, AUROC/Brier validation, and healing index score
8	Jul 20 — Storyboard	Results integration; storyboard and key figure drafts
9	Jul 27 — Poster draft	Poster design and layout
10	Jul 31 — Final poster	Final poster submission
11	Aug 11 — Final report	Write and submit final research report

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